VIKRANTH VAKATI, M.S.

Systems Engineer - U.S. Citizen

Boston, MA | vikvakati@gmail.com | linkedin.com/in/vikvakati | vikvakati.github.io/portfolio | (781)-363-4974

EXPERIENCE

Systems Engineer | Bodkin Design & Engineering | Newton, MA — May 2025 - Present

Electro-optical imaging solutions design for industrial and research communities

- Led development of software for system control and image processing using Python
- Collaborated in a cross-functional team to integrate software with hardware components
- Enhanced system performance by evaluating and implementing emerging technologies

Electrical Test Engineer | Delta Magnetics & Controls | Holbrook, MA — Jan 2021 - Jan 2023

Control panel design and fabrication for industrial process automation

- Wired and assembled control panels following industry standard IEC schematics
- Conducted visual and point-to-point testing to verify system functionality
- Deployed over 1000 control panels adhering to operational requirements

PROJECTS

Spacecraft Materials Testing | Embedded Software, Image Processing | Python — Apr 2027

Imaging system to characterize material degradation over time in a simulated space environment

- Produced wiring schematics compliant with class 3B laser safety requirements
- Designed software for synchronized motor control and image acquisition using multiple SDKs
- Implemented algorithm to generate BRDF measurements for use in 3D simulations

Autonomous Drone Cable Inspection | Embedded Software, Image Processing | Python — Jan 2027

Aerial imaging system to detect and monitor corrosion over time on galvanized steel guy wires

- Developed an image-based tracking algorithm to maintain precise alignment along a cable
- Leveraged sensor fusion to maintain stable tracking with dynamic cable conditions
- Improved inspection accuracy and repeatability, enabling long-term condition monitoring

Color Space Analysis Tool | Image Processing, UI Development | Python — Apr 2026

Interactive application to analyze image segmentation algorithms

- Implemented a GUI to convert between color spaces and tune parameters from live camera input
- Applied per channel thresholding, histogram equalization, and morphological filtering
- Streamlined selection of color spaces and threshold parameters to optimize image segmentation

Blackbody Calculator | Mathematical Modeling, Web Development | JavaScript, HTML/CSS — Jul 2025

Web tool to calculate and visualize spectral radiance and emittance | bit.ly/4uXHdIp

- Accurately formulated models based on first-principle physics laws
- Deployed a mobile-responsive UI with intuitive controls and real-time updates
- Improved response time compared to conventional tools through lazy data loading

EDUCATION

Wentworth Institute of Technology | Boston, MA

- M.S. in Computer Engineering, Concentration in Internet of Things — May 2024
- B.S. in Computer Engineering, Minors in Electrical Engineering and Applied Math — Dec 2022

SKILLS

Languages: Python, JavaScript, HTML/CSS, C/C++, Bash, MATLAB, SQL

Tools: VS Code, ImageJ, MAVLink, RESTful API, Windows, Linux, Excel, Git, LabView, MultiSim

Equipment: Multimeter, Oscilloscope, Spectral Cameras, Monochromator, Microcontrollers, SBCs

Communication Protocols: I2C, SPI, UART, RS232, RS485, Wi-Fi, Modbus, LoRa, RFID, BLE, Ethernet